

11(a)	packet/quantum of energy of electromagnetic/EM radiation	B1
11(b)(i)	$E = hf$ $1.1 \times 10^6 \times 1.60 \times 10^{-19} = 6.63 \times 10^{-34} \times f$	C1
	$f = 2.7 \times 10^{20} (2.65 \times 10^{20}) \text{ Hz}$	A1
11(b)(ii)	$p = h/\lambda = hf/c$ $= (6.63 \times 10^{-34} \times 2.65 \times 10^{20})/(3.00 \times 10^8)$ or $p = E/c$ $= (1.1 \times 1.60 \times 10^{-13})/(3.00 \times 10^8)$	C1
	$p = 5.9 \times 10^{-22} (5.87 \times 10^{-22}) \text{ N s}$	A1
11(c)	$123 \times 1.66 \times 10^{-27} \times v = 5.87 \times 10^{-22}$	C1
	$v = 2.9 \times 10^3 \text{ m s}^{-1}$	A1